

TLE:ICT – Computer Programming

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Description

This module is a subject-level learning package for Grade 9 or Grade 10 TLE-ICT Computer Programming. It is based on the Three-Term Budget of Work for Learning Competencies, where each TLE-ICT sector is studied for one term. The Computer Programming sector focuses on helping learners develop a website using HTML in a safe and responsible manner.

The module introduces learners to current programming trends, career opportunities, computer ergonomics, HTML structure, webpage content creation, multimedia, lists, tables, forms, links, CSS styling, animation, transition, box model, transform, and interactive website development. It gradually moves from basic awareness and HTML structure to a complete interactive website project.

Learning Objectives

By the end of this module, learners should be able to:

Discuss current trends, business/career opportunities, required skills in computer programming, and proper computer ergonomics. Identify and use basic HTML structure, elements, attributes, and HTML editors in creating simple webpages. Apply headings, paragraphs, styles, formatting, multimedia, lists, tables, forms, and links in HTML documents. Develop a simple multi-page website using HTML elements such as headings, paragraphs, images, lists, and links. Apply CSS syntax, selectors, colors, backgrounds, fonts, text styles, borders, margins, paddings, icons, links, lists, tables, animations, transitions, box model, shadow, and transform. Create an interactive webpage or website that responds to user actions using links, buttons, forms, multimedia, hover effects, and other interactive elements. Produce a final individual or group performance task by designing and building a functional, styled, interactive webpage or website using all HTML and CSS skills learned.

Lesson Overview

Lesson	Topic	Focus
Lesson 1	Programming Trends, Careers, and Computer Ergonomics	Learners explore programming careers, required skills, and safe computer workstation practices.
Lesson 2	HTML Structure, Elements, Attributes, and Editors	Learners create and edit basic HTML documents using proper structure and syntax.
Lesson 3	HTML Headings, Paragraphs, Styles, and Formatting	Learners design a simple information page using text structure and formatting.
Lesson 4	HTML Multimedia, Lists, and Tables	Learners add images/audio/video, create lists, and build simple tables in webpages.
Lesson 5	HTML Forms and Links	Learners create forms and apply hyperlinks to text, images, buttons, and email addresses.
Lesson 6	Basic Website Development	Learners plan and build a simple multi-page website using HTML elements.
Lesson 7	Introduction to CSS	Learners use CSS syntax, selectors, colors, backgrounds, fonts, and text styling.
Lesson 8	CSS Layout, Links, Lists, Tables, Animation, and Transition	Learners improve webpage appearance using spacing, styled links, icons, lists, tables, animation, and transition.
Lesson 9	CSS Box Model, Shadow, and Transform	Learners refine layout using margin, border, padding, box-shadow, and transform effects.
Lesson 10	Interactive Website Development	Learners create an interactive website using HTML and CSS skills, forms, buttons, links,

Lesson	Topic	Focus
		multimedia, and user interaction.

Programming Trends, Careers, and Ergonomics

Overview

Students discuss current programming trends, business and career opportunities, required skills, and proper computer ergonomics for safe and responsible computer use.

LESSON 1

Current Trends in Computer Programming

LESSON 2

Programming Careers and Required Skills

LESSON 3

Computer Ergonomics and Workplace Safety

HTML Structure and Editors

Overview

Students distinguish HTML structure, elements, and attributes, and use an HTML editor to create basic HTML documents.

LESSON 1

Creating HTML Documents Using an Editor

What We'll Learn

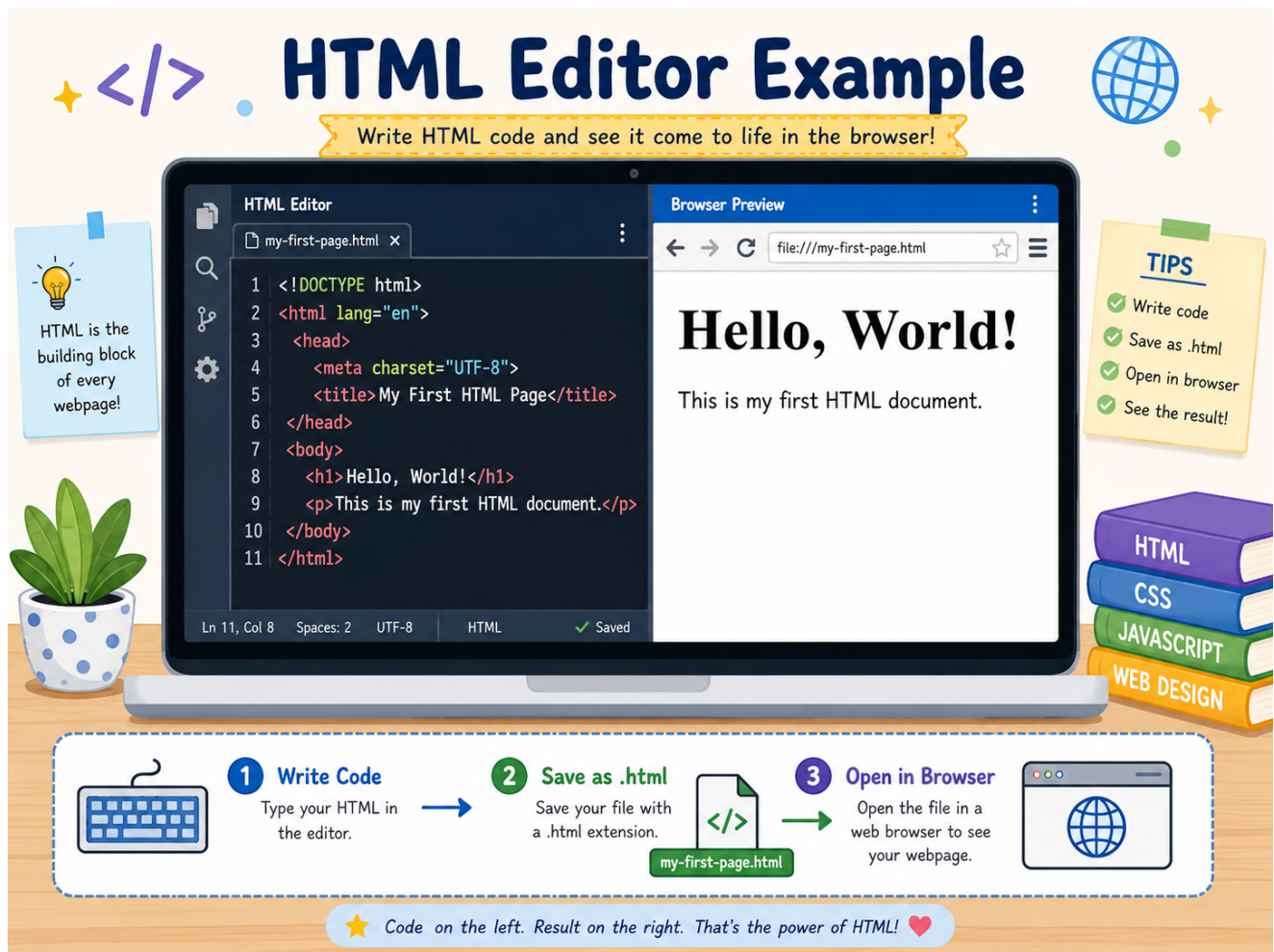
By the end of this lesson, students can use an HTML editor to create, save, and open a basic HTML document in a web browser.

Words We'll Use

- HTML Editor
- File
- Browser
- Save
- Preview

Before We Start

What apps or tools have you used to type or edit text on a computer?



An HTML editor is a tool used to write and edit HTML code. Examples include Notepad, Visual Studio Code, Sublime Text, or any simple text editor.

To create an HTML document, we write the HTML code, save the file using the .html extension, and open it in a web browser.

Example file name:

my-first-page.html

The browser reads the HTML file and displays the webpage.

A basic HTML document looks like this:

```
<!DOCTYPE html>
<html>
<head>
  <title>My First HTML Page</title>
</head>
<body>
  <h1>Hello, World!</h1>
  <p>This is my first HTML document.</p>
```

```
</body>
</html>
```

Let's Look at Examples

Example 1

HTML EDITOR EXAMPLE

Writing, Saving, and Viewing an HTML File

- Write HTML**
Type your HTML code in an editor.
- Save File**
Save the file with .html extension.
- Open in Browser**
Open the file in a web browser to see the result.

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>My First HTML Page</title>
5 </head>
6 <body>
7   <h1>Hello, World!</h1>
8   <p>This is my first HTML document.</p>
9 </body>
10 </html>
11
```

Remember: Save your file (with .html) and open it in a browser to see how it looks!

HTML ERROR FIXING EXAMPLE

Find the Errors and Fix Them

Incorrect Code (Has Errors)

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>About Me</title>
5 </head>
6 <body>
7   <h1>My Profile</h1>
8   <p>Hello! I am learning HTML.
9   
11 </html>
```

Corrected Code (Fixed)

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>About Me</title>
5 </head>
6 <body>
7   <h1>My Profile</h1>
8   <p>Hello! I am learning HTML.</p>
9   
10 </body>
11 </html>
```

Common HTML Errors

- Missing closing tags (e.g., </title>, </p>)
- Missing quotation marks in attributes
- Wrong or incomplete tags

Always Check:

- Opening and closing tags
- Correct order of tags
- Quotation marks in attributes

HTML STRUCTURE EXAMPLE

The Basic Structure of an HTML Document

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>My Web Page</title>
5 </head>
6 <body>
7   <h1>Welcome</h1>
8   <p>This is my first webpage.</p>
9 </body>
10 </html>
```

How it looks in the browser:

Example 1

Let's Practice

Open your HTML editor and create a new file.

Type the following HTML code:

```
<!DOCTYPE html>
<html>
<head>
  <title>My First HTML Page</title>
</head>
<body>
  <h1>Hello, World!</h1>
  <p>This is my first HTML document.</p>
</body>
</html>
```

Save the file as:

my-first-page.html

Open the file in a web browser and check if the heading and paragraph appear correctly.

Now We Apply

Create your own HTML document using an editor.

Your webpage should include:

- A page title
- One main heading
- One paragraph about yourself, your hobby, or your favorite subject

Save your file using the .html extension, then open it in a browser.

Challenge Task

Edit your webpage by adding:

- Another heading
- Another paragraph
- A short sentence explaining why you want to learn HTML

Save the file again and refresh the browser to see the changes.

How Our Work Will Be Checked

- File is saved with the .html extension
- HTML document opens correctly in a browser
- Page includes a title, heading, and paragraph
- Code uses the correct basic HTML structure
- Student can explain how to edit, save, and refresh the webpage

Let's Reflect

Why is it important to save the file correctly before opening it in a browser?

Resources / References

- HTML editor guide
 - images/html-editor-example.svg
-

LESSON 2

Fixing Basic HTML Syntax Errors

What We'll Learn

By the end of this lesson, students can identify and fix common HTML syntax errors in a simple webpage.

Words We'll Use

- Syntax
- Error
- Debug
- Opening Tag
- Closing Tag

Before We Start

Have you ever typed something incorrectly and the result looked different from what you expected?

HTML Error Fixing Example

With Errors	Fixed Version
1 <!DOCTYPE html>	1 <!DOCTYPE html>
2 <html>	2 <html>
3 <head>	3 <head>
4 <title> My Page	4 <title> My Page</title>
5 </head>	5 </head>
6 <body>	6 <body>
7 Welcome!	7 Welcome!
8 <p>This is a paragraph.	8 This is a paragraph.
9 	9
10 </body>	10 </body>
11 </html>	11 </html>

Errors in the code Corrected Fixed in the right version

HTML syntax is the correct way of writing HTML code. If the syntax is wrong, the webpage may not display correctly.

Common HTML errors include missing closing tags, wrong tag order, misspelled tags, missing quotation marks in attributes, and incorrect file extensions.

Example with errors:

```
<html>
<head>
<title>My Page</title>
<body>
<h1>Welcome</h1>
<p>This is my page.
</body>
</html>
```

Corrected version:

```
<html>
<head>
  <title>My Page</title>
</head>
```

```
<body>
  <h1>Welcome</h1>
  <p>This is my page.</p>
</body>
</html>
```

In the corrected version, the `<head>` section is closed properly and the paragraph has a closing `</p>` tag.

Let's Practice

Find and fix the errors in this HTML code:

```
<!DOCTYPE html>
<html>
<head>
  <title>About Me<title>
</head>
<body>
  <h1>My Profile</h1>
  <p>Hello! I am learning HTML.
  <img src=profile.jpg alt=Profile Picture>
</body>
</html>
```

Correct the code by checking the title tag, paragraph tag, and image attributes.

Now We Apply

Open a provided broken HTML file. Find the errors, correct them, save the file, and refresh the browser to check the result.

Your corrected webpage should:

- Have a proper HTML structure
- Use correct opening and closing tags
- Use quotation marks for attribute values
- Display properly in the browser

Challenge Task

Add comments beside the corrected lines explaining what you fixed.

Example:

```
<p>Hello!</p> <!-- Added missing closing paragraph tag -->
```

How Our Work Will Be Checked

- HTML errors are correctly identified
- Missing or incorrect tags are fixed
- Attributes are written properly
- Webpage displays correctly in the browser
- Corrections are explained clearly

Let's Reflect

What HTML error was easiest to fix? What error was hardest to find?

Resources / References

- HTML syntax guide
- [images/html-error-fixing-example.svg](#)